Safety-Critical Delays in Vehicular Networks Toward 6G: A Novel Metric for Disconnection Time Assessment

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***Abstract*—5G and emerging 6G V2X systems aim to enhance road safety through reliable, low-delay sidelink communications based on Semi-Persistent Scheduling (SPS) and Dynamic Scheduling (DS). Conventional metrics focus on packet-level reliability and overlook the temporal structure of outage events. We introduce a novel timeliness metric, Disconnection Time (DT), which quantifies the duration of V2X communication interruptions caused by consecutive packet losses, particularly persistent collisions in SPS. DT reveals prolonged outages that may have a limited impact on average inter-reception statistics, but may significantly affect safety-critical control loops. Simulation results show that, seen through the lenses of the new proposed metric, DS can outperform SPS, even for periodic traffic, especially at short inter-vehicle distances. These results highlight the importance of temporally aware metrics for future goal-oriented V2X and 6G standardization.**

***Index Terms*—5G, C-V2X, V2X, sidelink, semi-persistent scheduling, dynamic scheduling, persistence, vehicular networks, delay, disconnection time.**

I. Introduction

5G NEW Radio (NR)and beyond 5G solutions will play a dominant role in next-generation Intelligent Transportation Systems (ITS), supporting safer and more efficient mobility. Central to this evolution is Vehicle-to-Everything (V2X) communication, which extends situational awareness beyond onboard sensors to improve safety and decision-making. Within 5G systems, this capability is realized through New Radio (NR)-V2X, where sidelink communication enables direct vehicle-to-vehicle connectivity without relying on base stations [1].

Sidelink communication in 5G NR is used for status updates, like Cooperative Awareness Messages (CAMs), or to spread information about relevant detected objects with Collective Perception Messages (CPMs). In both cases, messages are broadcast from each vehicle to its neighbors (many-to-many traffic). This traffic pattern makes feedback unfeasible, as messages may be received by some vehicles, but missed by others. The lack of a feedback loop increases the risk of persistent collisions, as vehicles must autonomously manage transmission of awareness/perception messages based on sensing and timing, without the ability to adjust transmission parameters via feedback. In high-density environments, where disruptive interference (packet collision) is more likely, persistent collisions lead to prolonged information gaps. Consecutive losses of safety-critical messages can result in severe consequences, including delayed responses to warnings, hazards, or other critical events in autonomous and intelligent transportation systems [2].

Several standardized metrics are currently used to assess sidelink performance, including Packet Reception Ratio (PRR) [3], [4], Packet Inter-Reception Delay (PIR) [3], delay [4], and additional metrics such as Collision Loss Ratio (CLR) and Propagation Loss Ratio (PLR) [5], which quantify the likelihood of collisions and signal loss. Age of Information (AoI) and its variants [6], [7] are widely employed to evaluate the timeliness of status updates comprehensively. However, existing metrics do not thoroughly capture the underlying causes of AoI degradation [8], [9] or quantify the duration of collisions and complete disconnections between vehicles.

In this letter, we introduce a novel metric to quantify the duration of communication disconnections among vehicles caused by burst packet losses, such as persistent collisions. The letter focuses on the characterization of this safety-critical phenomenon through the lenses of the proposed new metric, providing insight for future 6G standardization targeting goal-oriented and semantic vehicular communications.

The contributions of this letter are summarized as follows:

- We introduce a new metric, named Disconnection Time (DT), to provide a detailed assessment of timeliness in sidelink communications.
- Based on the proposed metric, we highlight the impact of consecutive packet losses on information exchange among vehicles via an illustrative example (Section II) and detailed simulations of currently standardized access schemes namely Semi-Persistent Scheduling (SPS) and Dynamic Scheduling (DS).
- We show that for periodic message traffic flows SPS exhibits non-monotonic behavior, yielding DT performance worse than DS in high-density vehicular scenarios.
- Simulations results for DT highlight that information gaps are longer for SPS than with DS especially at small inter-vehicle distances, which exacerbates the safety issues brought about by persistent resource scheduling.

The rest of the letter is organized as follows. Section II discusses the impact on disconnections of persistent scheduling

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as defined in SPS for a simple, yet highly significant scenario. In Section III, we define novel metric DT and compare it against existing standardized metrics. Section IV details the simulation setup, while Section V presents simulation results. Finally, conclusions are drawn in Section VI.

II. Resource Allocation and Persistent Collisions

The sidelink channel is organized into Sub-Channels (SCs), which are the resources selected by nodes that need to transmit a message. In the following, we assume that an update message fits into a single SC. The key features of the SPS procedure standardized in [10] and [11] are summarized in Section II-A, while DS is described in Section II-B. Persistent scheduling results in periodic reservation and hence usage (if a message is available) of the same radio resources. The effect of persistence on persistent collisions, for a special, yet highly significant case, is investigated in Section II-C.

A. Semi-Persistent Scheduling

SPS is designed to match periodic traffic patterns. The key idea of SPS is to keep a selected SC for a number of consecutive times, according to a periodic reservation pattern, with period Resource Reservation Interval (RRI). The SPS mechanism relies on two parameters, i.e., the Reselection Counter (RC), that defines the number of consecutive uses of the same SC, and the persistence probability (P), set to a value in the range $[0, 0.8]$. Once a new SC is selected, a value of RC is drawn uniformly at random in an integer set (see later on for details). Let k be the value of the randomly chosen RC. After having kept the selected SC k times consecutively, with period RRI, a new value k' of the RC is drawn and the same SC is retained further for k' times with probability P. On the contrary, with probability $1-P$, a new SC is selected, different from the one currently used.

The initial value of the RC depends on the selected RRI. It is an integer, drawn uniformly at random in the interval $\{\lceil 5{\cdot}C\rceil, \ldots, \lceil 15{\cdot}C\rceil\}$.[1] The quantity C is computed as follows:

$$C = \max\left\{1, \min\left\{5, \frac{\text{RRI}_{\text{th}}}{\text{RRI}}\right\}\right\} \tag{1}$$

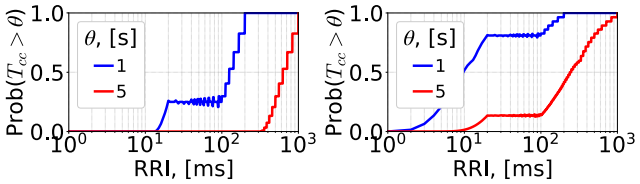
where $\text{RRI}_{\text{th}} = 100\text{ms}$. Let us introduce the following notation:

$$r_l = \lceil 5 \cdot C\rceil, \quad r_u = \lceil 15 \cdot C\rceil, \quad \overline{r} = 10 \cdot C \tag{2}$$

Then, the initial value of the RC is drawn at random in the integer set $\{r_l, \ldots, r_u\}$, while $\overline{r}$ denotes the mean value of the RC. Note that $C \geq 1$, hence $\overline{r} \geq 10$.

The number of times that the same SC is kept with period RRI is the realization of a random variable N. The probability distribution of N is found by considering that $N = \sum_{j=1}^{K} U_j$, where K is a Geometric random variable with $\mathcal{P}(K = h) = (1-P)P^{h-1}$, $h \geq 1$, and U_j are independent and identically distributed (i.i.d.) uniform random variables drawn from the integer set $\{r_l, \ldots, r_u\}$. Then, we have

[1] $\lceil x\rceil$ denotes the ceiling function of x, i.e., the smallest integer not smaller than x.



(a) $P = 0$ (b) $P = 0.8$

Fig. 1. Probability that persistent collision time T_{cc} exceeds threshold $\theta = 1$ sec and $\theta = 5$ sec as a function of RRI, for three values of the persistence probability P.

$$\mathcal{P}(N \geq n) = \begin{cases} \mathcal{P}(U_1 \geq n) & \text{w.p. } 1-P, \\ \mathcal{P}(\tilde{N} \geq n - U_1) & \text{w.p. } P. \end{cases} \tag{3}$$

where $\tilde{N}$ is the residual number of reservations of the same SC. Thanks to the memoryless property of the Geometric random variable, it is $\tilde{N} \sim N$. Let $G_N(n) = \mathcal{P}(N \geq n)$ and $G_U(n) = \mathcal{P}(U \geq n)$. Since U is uniform over $\{r_l, \ldots, r_u\}$, for any integer n it is

$$G_U(n) = \min\left\{1, \max\left\{0, \frac{r_u + 1 - n}{r_u + 1 - r_l}\right\}\right\} \tag{4}$$

Then, for all $n \geq 1$ we have

$$G_N(n) = (1-P)\, G_U(n) + \frac{P}{r_u + 1 - r_l} \sum_{k=r_l}^{r_u} G_N(n-k) \tag{5}$$

with the convention that $G_N(n) = 1, \forall n \leq 0$. Using Equations (4) and (5), the probability distribution of N can be computed efficiently.

B. Dynamic Scheduling

The DS resource allocation mechanism in 5G NR-V2X, as defined in [10] and [11], allows a node to dynamically select a new resource for each transmission, without persistence. After each transmission, the node re-selects a random SC among those that are sensed to be idle in the selection window. In other words, DS is the reservation-less variant of the SPS scheme which sets $RC = 1$ and $P = 0$. Compared to SPS, DS may incur more collisions with respect to SPS, thus reducing the achieved PRR level especially under periodic traffic. Yet, packet losses are scattered randomly over time with DS, thus reducing drastically the probability of persistent collisions with respect to SPS. Studies show that under high channel load [8] or aperiodic traffic [12], [13], DS can outperform SPS while simplifying the resource allocation algorithm.

C. Persistent Collisions for Two Nodes

Let us consider two nodes, A and B, selecting the same SC with the same RRI. In the following we assume greedy nodes, i.e., each reserved SC is actually used for transmitting a new update message. Other nodes that receive the signals of both A and B will likely experience excessive interference, thus failing to detect either message (collision event). As a consequence, all nodes experiencing a collision will lose track of nodes A and B, given the impossibility of receiving a successful update message as long as A and B keep using

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TABLE I
QUANTILES OF T_{cc} FOR DIFFERENT P VALUES AND RRI SETTINGS

P = 0	**RRI = 20 ms**	**50 ms**	**100 ms**	**200 ms**
Quantile 90 of T_{cc} (s)	0.71	1.42	1.50	3.00
Quantile 99 of T_{cc} (s)	0.76	1.52	1.60	3.20
P = 0.5	**RRI = 20 ms**	**50 ms**	**100 ms**	**200 ms**
Quantile 90 of T_{cc} (s)	1.97	3.94	4.00	8.00
Quantile 99 of T_{cc} (s)	3.68	7.36	7.50	15.00
P = 0.8	**RRI = 20 ms**	**50 ms**	**100 ms**	**200 ms**
Quantile 90 of T_{cc} (s)	5.45	10.90	11.00	22.00
Quantile 99 of T_{cc} (s)	10.66	21.32	21.40	42.80

the same SC.[2] Even if collision events may be rare (e.g., one collision events every 1000 messages), losing several messages consecutively may result in "blinding" times of the order of seconds, depending on parameter configuration of SPS, which is definitely a safety-critical issue.

Let N_A and N_B denote the number of consecutive times that the selected SC is used with the same RRI by A and B, respectively. Under SPS, we have $N_A, N_B \sim N$. Let N_{cc} be the random variable defined as the number of consecutive collisions between A and B. Then, it is $N_{cc} = \min\{N_A, N_B\}$. Since N_A and N_B are independent random variables, we have $\mathcal{P}(N_{cc} \geq n) = \mathcal{P}(\min\{N_A, N_B\} \geq n) = [G_N(n)]^2$.

The duration of the persistent collision is $T_{cc} = R \cdot N_{cc}$, where R is the common RRI used by A and B. The probability that the persistent collision time T_{cc} exceeds a threshold θ is therefore $\mathcal{P}(N_{cc} \geq \lceil \theta/R \rceil) = [G_N(\lceil \theta/R \rceil)]^2$.

Section II-C shows the probability that the persistent collision time T_{cc} exceeds threshold $\theta = 1$ s and $\theta = 5$ s as a function of RRI. Two values of the persistence probability are considered, namely $P = 0, 0.8$, corresponding to the two extreme values allowed for P in SPS. The non-trivial behavior of the curves depends on the fact that the probability distribution of T_{cc} depends on the value of RRI both directly and through the parameter C defined in Equation (1).

It is apparent that, as RRI grows beyond a few tens of ms, the probability that nodes A and B are blinded to their neighbors because of persistent collisions becomes substantial. As a matter of example, for values of RRI ranging between 20 ms and 100 ms, the probability of a persistent collision time of more than 1 sec is ≈ 0.8 for $P = 0.8$, and ≈ 0.25 for $P = 0$. A persistent collision time of 5 sec is exceeded with probability 0.17 for $P = 0.8$.

Further evidence that long collision times are possible with non-negligible probability is provided by the results presented in Table I, which collects 90 and 99 quantiles of T_{cc}. With $P = 0.8$, a persistent collision time of 11 sec is exceeded with probability 0.1 for RRI equal to 100 ms.

Note that Section II-C and Table I plots the *conditional* probability of a long collision time, given that A and B enter into a collision event by selecting the same SC. The probability of a collision event is not necessarily small, as shown in extensive studies of SPS performance (e.g., see [8], [14]). Once a collision occurs, the results in Section II-C and Table I show that A and B become invisible to neighboring nodes for quite a long time with high probability, unless RRI is very small and update messages are actually transmitted at high rate, which however may affect adversely collision probability. It therefore appears that the persistence built into SPS can pose a safety hazard, since missing updates from nearby vehicles for several seconds severely affects the achievable level of safety in assisted and autonomous driving applications.

Albeit a simplistic scenario, two vehicles example allows a crisp quantitative understanding of the issue of persistent collisions as a function of SPS parameters. In next Section, the impact of prolonged collisions is evaluated in a realistic scenario, where multiple vehicle share the sidelink channel.

III. NEW METRIC DEFINITION

The DT quantifies the total time during which a receiver does not successfully receive any information from a transmitting node. The disconnection time $D_{ij}(t)$ of node j with respect to a transmitter node i is defined as the time elapsing between two consecutive successful receptions at j from i, *provided that at least one message transmitted by i has been lost to j in between*. Two remarks are in order.

First, DT is not equivalent to other standardized metrics, and specifically to PIR. Consider a sample of m inter-reception intervals $I_1, I_2, \ldots, I_m$ for messages transmitted by node i and successfully received by node j. Each interval I_h represents the time between two consecutive successful receptions at node j. Let also $I_{j_1}, \ldots, I_{j_k}$, with $k \leq m$, denote the subset of intervals associated with non-consecutive transmissions, that is, intervals containing at least one packet loss. Average PIR_{ij} from node i to node j is estimated as:

$$\overline{PIR}_{ij} = \frac{1}{m} \sum_{h=1}^{m} I_h \tag{6}$$

Average DT_{ij} is evaluated as:

$$\overline{DT}_{ij} = \frac{1}{k} \sum_{\ell=1}^{k} I_{j_\ell} \tag{7}$$

which is clearly different from $\overline{PIR}_{ij}$. The fundamental distinction is that, while PIR quantify the elapsed time between successful receptions, DT isolates only those intervals where packet losses occur.

In practice, when PRR is high, say 0.9, $\overline{PIR}_{ij}$ is dominated by short intervals, equal to one RRI under the greedy node assumption, so that longer gaps caused by consecutive packet losses are averaged out. In contrast, $\overline{DT}_{ij}$ is computed only from intervals involving at least one packet loss, therefore directly characterizing the duration of communication outages. This difference is illustrated in Section III, which depicts the temporal evolution of PIR, AoI, and DT. Messages are generated at times g_k and received at times a_k, for $k \in \{1, \ldots, 11\}$, with successful receptions marked in green and losses in red. The AoI plot exhibits the characteristic sawtooth behavior, while PIR corresponds to the inter-reception intervals $I_n = a_n - a_{n-1}$. The DT metric, by restricting attention to

[2] A fix to this issue has been proposed in SAE J2945, the so called one-shot transmission. Note that this fix inflates the number of SCs reserved by a node, for the same message traffic. We do not consider this feature here.

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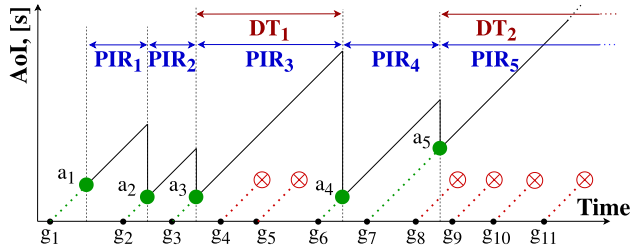


Fig. 2. Illustrative example of PIR, AoI and DT comparison.

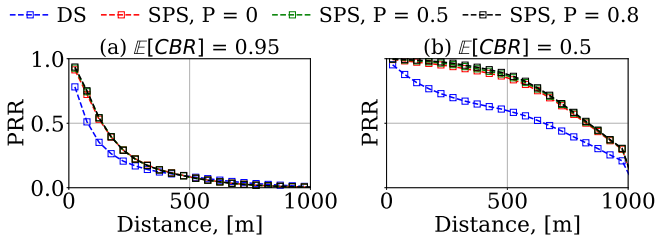
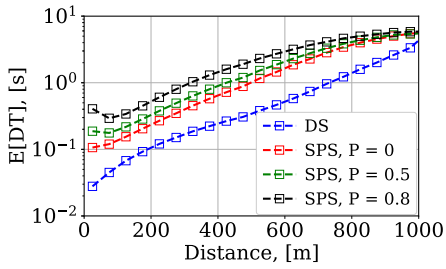
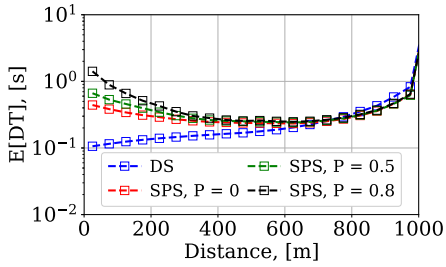


Fig. 3. PRR for (a) high and (b) low channel loads as a function of transmitter-receiver distance for different configurations of SPS and DS.



(a) $\mathbb{E}[\mathrm{CBR}] = 0.95$



(b) $\mathbb{E}[\mathrm{CBR}] = 0.5$

Fig. 4. DT for (a) high and (b) low channel loads as a function of transmitter-receiver distance for different configurations of SPS and DS.

intervals containing losses, isolates persistent outage periods that may only marginally affect the average PIR.

Second, any node can estimate DT statistics autonomously. The receiving node j simply collects samples of PIR_{ij} from node i and isolates those PIR intervals that exceed the RRI declared by the transmitting node i in the Sidelink Control Information (SCI) of the SC where the transmission was made.

In assisted and autonomous driving applications, DT can be interpreted as a real-time confidence indicator for incoming V2V information. It quantifies the duration of blind periods affecting communications with neighboring vehicles, without requiring explicit feedback. By highlighting information gap intervals, DT provides an assessment of temporal reliability beyond standard PIR or even AoI analysis. Monitoring the disconnection time of nearby nodes enables adaptive control actions, such as increasing inter-vehicle headway when DT exceeds predefined safety thresholds.

From a system design perspective, DT supports MAC-layer parameter tuning, e.g., persistence probability in SPS. Persistent scheduling may create highly correlated transmission patterns, leading to prolonged disrupting interference between close-by vehicles. As discussed in Section V, DT reveals that vehicles separated by short distances can experience extended blind periods due to correlated sensing and persistence.

IV. Simulation Setup

We simulate a highway scenario with a 5000 m stretch, three lanes in each direction, and vehicles traveling at 70 km/h^{-1}. The minimum safe headway is 40 m, yielding a maximum vehicle density of 150 veh km $^{-1}$ (as derived in [8]). The average vehicle density in simulations is set to 140 veh km^{-1}. We evaluate two channel load scenarios based on the Channel Busy Ratio (CBR) (ratio of occupied SCs over 100 subframes [1]): low channel load ($\mathbb{E}[\mathrm{CBR}] = 0.5$) and high channel load ($\mathbb{E}[\mathrm{CBR}] = 0.95$). For the considered average vehicle density, the two scenarios area obtained by adjusting the generation rate of update messages, namely $T_{\mathrm{gen}} = \mathrm{RRI} = 100$ ms for the low CBR regime and $T_{\mathrm{gen}} = \mathrm{RRI} = 20$ ms in case of high CBR.[3]

The MoReV2X simulation tool [5], based on ns-3, models vehicle mobility, propagation, and multiple access, focusing on sub-6 GHz NR-V2X communications. It uses Mode 2 sidelink for direct V2V communication, with DS and SPS mechanisms.

[3]In general, not all reserved SCs must necessarily be used for transmissions. Hence, a lower RRI does not necessarily imply a higher channel usage. In this letter, we assume greedy nodes, so a new message is transmitted in each reserved SC.

The OFDM numerology is set to 0, with a Sub-Carrier Spacing (SCS) of 15 kHz, slot duration of 1 ms, and NR-V2X radios operating on a 10 MHz channel in the 5.9 GHz ITS band. 10 Resource Blocks (RBs) are used per SC (in total 5 SCs per slot), with Modulation and Coding Scheme (MCS) 13 (16-QAM, 490/1024 code rate), allowing for the transmission of a 299-byte packet. Transmit power is set to 23 dBm, and the Received Signal Strength Indicator (RSSI) threshold is -92.3 dBm. Path loss and shadowing models follow [3], and Block-Error Rate (BLER) curves from [15] model fast-fading impairments. The Signal-to-Noise-plus-Interference Ratio (SNIR) value determines the BLER for successful message reception [15]. As for sensing and SC selection, the initial Reference Signal Receive Power (RSRP) threshold is -120.27 dBm with a selection window fraction $\beta = 20$ %. Simulations assess DS and SPS with persistence probabilities of 0, 0.5, and 0.8 to span the standardized SPS performance spectrum. System parameters are summarized in Table II.

V. Numerical Results

This section evaluates the impact of distance and channel load on reliability and timeliness by jointly analyzing PRR and DT under SPS and DS. PRR measures the probability of suc-

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TABLE II
NUMERICAL VALUES OF MAIN SYSTEM PARAMETERS

Parameter	Values
Highway length, L	5000 m
Vehicle density, ρ	140 veh/km
Number of lanes	6
Vehicle speed	70 km/h
RRI	20 and 100 ms
Message generation time, T_{gen}	20 and 100 ms
Channel bandwidth, BW	10 MHz
Sub-Carrier Spacing, B_{sc}	15 kHz
Time slot duration, T_s	1 ms
MCS	MCS-13
# of RBs per Sub-Channel	10
# of Sub-Channels per time slot, n_{SC}	5
Noise figure	9 dB
Margin for idle/busy channel assessment	3 dB
Probability of persistence, P	0; 0.5; 0.8

cessful packet reception [3], from which the loss probability can be derived, whereas DT captures the duration of outages, due to consecutive packet losses.

The PRR, averaged across all vehicles, is plotted in Section 3 as a function of distance between transmitter and receiver for DS and for three values of persistence probability in case of SPS. PRR decreases rapidly with distance under high channel load, while it stays lose to 1, at least for short distances, in case of low load. It is noted that, even with PRR ≈ 0.99, an average disconnection time of 1 s corresponds, on average, to a 1 s outage every 100 s. Despite their low frequency, such outages may critically affect safety in close-range interactions. Section V illustrates average DT as a function of distance between transmitter and receiver for DS and for three values of persistence probability in case of SPS. It is apparent that outages are shorter with DS in both high and low load regimes. Random resource reservation leads to lower PRR, but it breaks correlations created by periodic reservation, thus avoiding long gaps. For high channel load, DT grow with distance, except that there is a trend inversion at short distances for high persistence levels. The non monotonic behavior is more pronounced in case of low load regime, where DT is definitely non monotonic in case of SPS, with a minimum between 400 m and 600 m.

The left-tail observed for DT in case of SPS at short distances is rooted in packet collisions rather than propagation loss. Vehicles in close proximity observe nearly identical channel states, leading to highly correlated SC selection. This "synchronization" increases collisions, which high persistence levels (P) fail to resolve quickly. More distant vehicles are loosely coupled by sensing, hence their transmissions may overlap only partially upon persistent reservation of a same SC. Notably, the left-tail disappears with DS, as random, non-persistent SC selection breaks the "collision trap," limiting the duration of any collision event.

Finally, average DT is larger in case of low load, at short distances, since RRI equals 100 ms in that case, five times greater than in case of high load. This highlights that DT analysis must account for both CBR levels and message rate.

VI. CONCLUSION

This study highlights the importance of new metric, the Disconnection Time, in evaluating the performance of 5G and emerging 6G V2X communications. This metric provide a deeper understanding of the *continuity* of update message flows in such systems, offering additional insight into the impact of persistence on communication reliability and timeliness.

Two main results stem from the analysis of DT: (i) long blind periods (in the order of seconds) are possible, with non-negligible probability, under SPS, the more the higher the persistence probability; (ii) close-by vehicles (at distance in the order of few tens of meters) may incur in long blind periods under SPS, because of persistence and of the high correlation of sensing. Both effects disappear with DS, which turns out to offer better performance in terms of disconnection times as compared to SPS.

The results in this letter may be useful to enhance characterization of multiple access schemes in NR-V2X sidelink and future 6G-based standards, with the goal of designing simple yet effective mechanisms that ensure high reliability and timely information dissemination, particularly in safety-critical scenarios, such as in proximity to other vehicles and areas of high vehicle density.

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